

Informed Melody Extraction from Symbolic Data

AIST 3310 Project Presentation

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Introduction

Goal: Identify the melodies of a polyphonic music piece with heuristics.

Current Approaches:

- Skyline: Find Highest Pitch (G. Ozcan, C. Isikhan, A. Alpkocak, 2005)

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Current Approaches:

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- Deep Learning (Y. Chou, I. Chen, C. Chang, J. Ching and Y. Yang, 2021)



Figure 1. Illustration of our baselines. a) Input example, b) Output of skyline algorithm, c) Output of skyline variation algorithm.

Neural Network

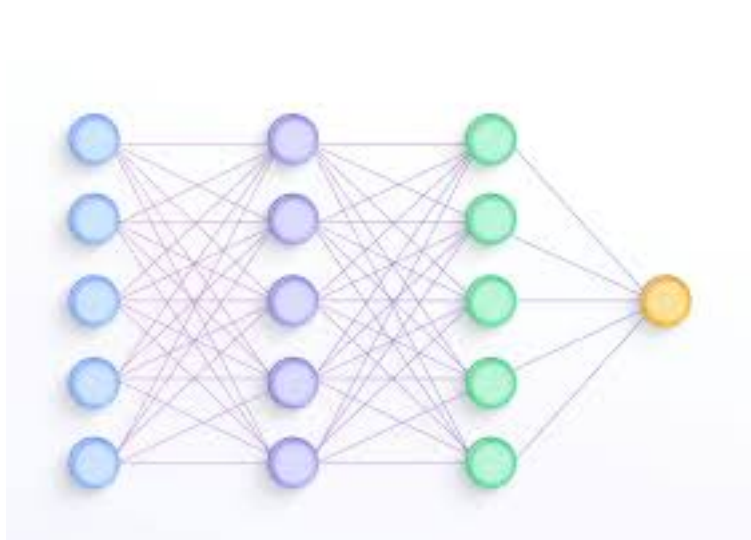


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Heuristics Identified

Pre-processing

Remove all chords and all percussion first.

- Loudest Dynamic
- Highest Pitch
- Significant Pitch Delta
- Short Rhythm
- Articulation Marks
- Continuity

With these heuristics we are able to assign a score to each of the measures.

Example

B

solo

f

The image displays a musical score for a solo section, labeled 'B'. The score is written on four staves. The first staff begins with a forte (*f*) dynamic marking and a 'solo' instruction. It contains a melodic line starting with a quarter rest, followed by a series of eighth notes and a dotted quarter note. The second staff is empty. The third staff contains a melodic line starting with a quarter rest, followed by a series of eighth notes and a dotted quarter note. The fourth staff is empty. The score includes various musical notations such as notes, rests, and dynamic markings.

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Psuedocode

```
k := number of instruments we keep at each measure
W := weights for each heuristic
S := score for each instrument at each measure
R := new score
for each measure in piece:
    for each instrument in measure:
        X := extract_feature()
        S.append(W @ S)
    K := symbolic data of the top k instruments
    R.append(K)

return R
```

Feature Extraction

We want to normalize each of the heuristics to a 0-1 value.

Recall that articulation, rhythm shortness and continuity are already normalized as they are True/False values.

Thus, we can either normalize with a discrete continuous function.

$$f(\text{pitch}) = \begin{cases} 1 & \text{if pitch} < \text{mode or pitch} = \text{max} \\ 0 & \text{otherwise} \end{cases}$$

Or

$$f(p\delta_i) = \frac{p\delta_i}{\text{argmax}_{instruments}(p\delta)}$$

Training

We use Multiplicative Weight Update method, a method widely used in Machine Learning and Game Theory, to train our weights.

We update the weight of each wrong heuristic w_i at each iteration to be

$$w_i^{t+1} = w_i^t(1 - \lambda)$$

where λ is a fixed constant. We then normalize all the weights such that they sum to 1.

This gives us a guarantee in expectation polynomially bound by m^* minimum error and n number of heuristics. (S. Arora, E. Hazan, S. Kale, 2012)

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Results

We tested this algorithm on a section Dvorak's New World Symphony Movement IV. The accuracy score for our method is filled in from this test.

Heuristic	Skyline	CNN
88.9%	78.7%	91.7%

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We find that while such an expert system is generally better than the single heuristic Skyline, it has the following weaknesses.

- Missing Pickups
- Long Ending Notes
- Counter-melody and "frill" notes
- Absolute Volume

Additionally, some finer grained control could be introduced in place of coarse heuristics such as pitch delta. Some additional heuristics such as sectional similarity can also be employed.

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References

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